

The empirical recurrence for OEIS A224969: recovering a conjecture that is not written down

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Abstract

OEIS A224969 records “Empirical recurrence of order 80”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 318$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 80, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 80 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A224969 is “Number of $(n+1) \times 3$ 0..3 matrices with each 2×2 permanent equal.”. It has offset 1 and begins

912, 11196, 110253, 1777586, 26732072, 435788333, 6995454550, 113660299418, ...

The entry states:

Empirical recurrence of order 80 (see link above)

As of the “Last modified” line on the live entry (Oct 03 2025 23:37:10, revision 9) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 318$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 318$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 636$ exact terms of the model returns order 80 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{80} c_i a(n-i),$$

c_1	36	c_{21}	−111450814700640205	c_{41}	3426781260292709154826893	c_{61}	−232
c_2	−260	c_{22}	687893649123868300	c_{42}	−3961069827554630030114184	c_{62}	1489
c_3	−4653	c_{23}	587891755496820654	c_{43}	−6543959000043947341864718	c_{63}	698
c_4	70515	c_{24}	−7250557785349530556	c_{44}	11385289732117559017461629	c_{64}	−86
c_5	112369	c_{25}	429592132368370909	c_{45}	9096871410523469157360295	c_{65}	−29
c_6	−6501642	c_{26}	60148885263594448516	c_{46}	−25332160282998248344533130	c_{66}	318
c_7	16117645	c_{27}	−46673819936550790113	c_{47}	−6883244342628662709818272	c_{67}	−81
c_8	326603854	c_{28}	−395919185601798727827	c_{48}	44638750660274105623903773	c_{68}	−68
c_9	−1682301482	c_{29}	568580673877522684628	c_{49}	−5230206508476015454470257	c_{69}	397
c_{10}	−9826765907	c_{30}	2055886367052024440314	c_{50}	−62292458908679755289660702	c_{70}	41
c_{11}	83118236933	c_{31}	−4464133893904401788446	c_{51}	28317237879308160765980928	c_{71}	−8
c_{12}	158915223315	c_{32}	−8187991084856088754525	c_{52}	67476030137508808544736678	c_{72}	17
c_{13}	−2663354677076	c_{33}	26358512578233638575394	c_{53}	−54799579221857010528951755	c_{73}	78
c_{14}	193518754525	c_{34}	22840239859144321902036	c_{54}	−53629843325725399026343960	c_{74}	−4
c_{15}	60647463977450	c_{35}	−122950316139496129027103	c_{55}	70940939890500886858657133	c_{75}	3
c_{16}	−86501178877639	c_{36}	−27573083486850372345804	c_{56}	25965963415243115624243781	c_{76}	2
c_{17}	−1017078139718781	c_{37}	461882780507105953176757	c_{57}	−67245696927810889553732598	c_{77}	−
c_{18}	2656810161656074	c_{38}	−124142093911788311471348	c_{58}	925059610047778627667333	c_{78}	1
c_{19}	12615508738391340	c_{39}	−1403950864428638162124418	c_{59}	47042080452316631887522036	c_{79}	
c_{20}	−50150569657666225	c_{40}	988134741723531010397079	c_{60}	−14848373917887064242424439	c_{80}	

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 80, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $80 + 4$ terms removed returns order 80 as well, so $\deg q_{\min} = 80 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $80 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 80 that this sequence satisfies — and that family turns out to have exactly one member.

References

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